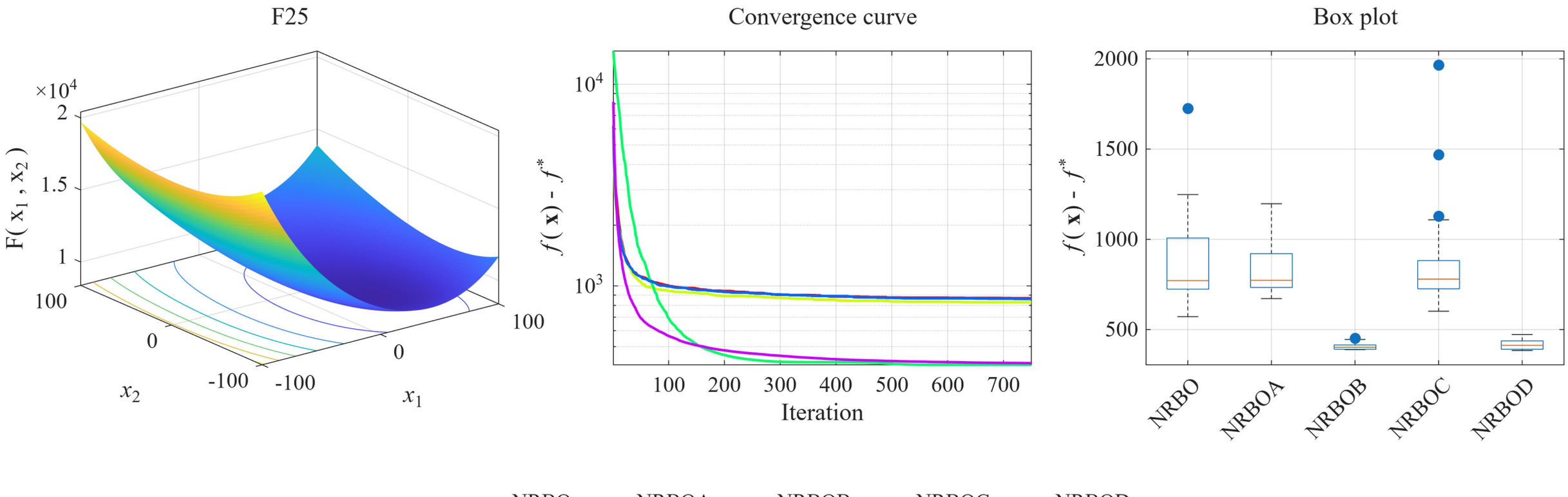

F25
×10^4
2
1.5
1
F(x1 , x2)
100
0
-100
x2
-100
0
100
x1
Convergence curve
10^4
10^3
f(x) - f*
100
200
300
400
500
600
700
Iteration
Box plot
2000
1500
1000
500
f(x) - f*
NRBO
NRBOA
NRBOB
NRBOC
NRBOD
NRBO
NRBOA
NRBOB
NRBOC
NRBOD